180: Counting Techniques

In the following exercise we demonstrate the use of a few fundamental counting principles, namely the addition, multiplication, complementary, and inclusion-exclusion principles. While none of the principles are particular complicated in their own right, it does take some practice to become familiar with them, and recognise when they are applicable. I have attempted to indicate where alternate approaches are possible (and reasonable).

Problem: Assume $n \geq 2$ and $m \geq 1$. Count the number of functions $f : [n] \to [m]$

(i) in total.

(ii) such that $f(1) = 1$ or $f(2) = 1$.

(iii) with $\min_{x\in[n]} f(x) \leq 5$.

(iv) such that $f(1) \geq f(2)$.

(v) that are strictly increasing; that is, whenever $x < y$, $f(x) < f(y)$.

(vi) that are one-to-one (injective).

(vii) that are onto (surjective).

(viii) that are bijections.

(ix) such that $f(x) + f(y)$ is even for every $x, y \in [n]$.

(x) with $\max_{x\in[n]} f(x) = \min_{x\in[n]} f(x) + 1$.

Solution:

(i) A function $f : [n] \to [m]$ assigns for every integer $1 \leq x \leq n$ an integer $1 \leq f(x) \leq m$. For each integer x, we have m options. As we make n such choices (independently), the total number of functions is $m \times m \times \ldots m = m^n$.

(ii) (We assume $n \geq 2$.) Let A_1 be the set of functions with $f(1) = 1$, and A_2 the set of functions with $f(2) = 1$. Then $A_1 \cup A_2$ represents those functions with $f(1) = 1$ or $f(2) = 1$, which is precisely what we need to count. We have $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$.

If $f(1) = 1$, then we have only 1 option for $f(1)$, and m options for $f(x)$, $2 \leq x \leq n$. Thus $|A_1| = 1 \times m \times \ldots \times m = m^{n-1}$. Similarly, $|A_2| = m^{n-1}$.

Now we count $|A_1 \cap A_2|$. These are the functions for which $f(1) = 1$ and $f(2) = 1$. Thus there is only 1 option for both $f(1)$ and $f(2)$, and m options for $f(x)$ with $3 \leq x \leq n$. Thus $|A_1 \cap A_2| = 1 \times 1 \times m \times \ldots \times m = m^{n-2}$.

Putting this together, we have $|A_1 \cup A_2| = m^{n-1} + m^{n-1} - m^{n-2} = (2m - 1)m^{n-2}$.

Note: An alternative solution is to consider the complement instead - count those functions that do not satisfy the given property, and then subtract them from the total number of functions. In this case, the complement consists of those functions for which $f(1) \neq 1$ and $f(2) \neq 1$. Thus there are $m - 1$ choices for both $f(1)$ and $f(2)$, and m choices for $f(x)$ with $3 \leq x \leq n$. Thus the number of 'bad' functions is $(m - 1) \times (m - 1) \times m \times \ldots \times m = (m - 1)^2 m^{n-2}$. Hence the number of 'good' functions is $m^n - (m - 1)^2 m^{n-2} = (m^2 - (m - 1)^2)m^{n-2} = (2m - 1)m^{n-2}$.

(iii) If $m \leq 5$, then $f(x) \leq m \leq 5$ for all $1 \leq x \leq n$, and so every function satisfies $\min_{x\in[n]} f(x) \leq 5$. Hence the number of such functions will be m^n. Now we will assume that $m > 5$.

We solve this problem by counting the complement. The 'bad' functions will be those with $\min_{x\in[n]} f(x) > 5$. These are precisely the functions for which $6 \leq f(x) \leq m$ for every $1 \leq x \leq n$. For each x, we have $m - 5$ options, and so the number of 'bad' functions is $(m - 5)^n$. Thus the number of 'good' functions - those with $\min_{x\in[n]} f(x) \leq 5$ - is $m^n - (m - 5)^n$.

(iv) We split this problem into m cases, depending on the value of $f(1)$. If $f(1) = k$, for any $1 \leq k \leq m$, then we know that $1 \leq f(2) \leq k$, and so there are k choices for $f(2)$. For the remaining values $f(x)$, $3 \leq x \leq n$, there are no restrictions, and so each value has m options. Thus, if $f(1) = k$, there are $k \times m \times \ldots \times m = km^{n-2}$ ways to extend the function to all of $[n]$.

The possible values for k are $1, 2, \ldots, m$. Thus the total number of functions with $f(1) \geq f(2)$ is given by

$$\sum_{k=1}^{m} km^{n-2} = \left(\sum_{k=1}^{m} k\right) m^{n-2} = \frac{m(m+1)}{2} m^{n-2} = \frac{1}{2} m^{n-1}(m+1).$$

Note: Alternatively, one can split into the cases where $f(1) = f(2)$ and $f(1) > f(2)$. If $f(1) = f(2)$, there are m options for $f(1)$, and only 1 option for $f(2)$, and m options for all the other values, giving m^{n-1} such functions. To count the functions with $f(1) > f(2)$, we first count those with $f(1) \neq f(2)$. As there are m^n functions, and m^{n-1} functions with $f(1) = f(2)$, there are $m^n - m^{n-1} = m^{n-1}(m-1)$ functions with $f(1) \neq f(2)$. We claim that exactly half of these functions have $f(1) > f(2)$. Indeed, we can take any function with $f(1) \neq f(2)$, and then swap $f(1)$ and $f(2)$ to get a new function g. If $f(1) > f(2)$, then $g(1) < g(2)$, and if $f(1) < f(2)$, then $g(1) > g(2)$. Thus there are an equal number of functions with $f(1) > f(2)$ as with $f(1) < f(2)$, and thus the number of functions with $f(1) > f(2)$ is half the number of functions with $f(1) \neq f(2)$, or $\frac{1}{2} m^{n-1}(m-1)$. Adding the two cases, we get that the number of functions with $f(1) \geq f(2)$ is $m^{n-1} + \frac{1}{2} m^{n-1}(m-1) = \frac{1}{2} m^{n-1}(m+1)$.

(v) The key is to note that since no two of the values $f(x)$ can be the same, the image of a strictly increasing function f must have n elements. Moreover, if you specify n elements to be the image of a strictly increasing function f, there is only one possibility for the function, as the image must appear in increasing order. Thus the number of strictly increasing functions is precisely the number of ways of selecting n elements from the m numbers in the range $[m]$, and is thus $\binom{m}{n}$.

(vi) Note that if $n > m$, then there must be two different integers $x \neq y$ with $f(x) = f(y)$ (this is, formally, the pigeonhole principle), and so there cannot be a one-to-one function from $[n]$ to $[m]$ in this case. We will assume $n \leq m$ in the argument that follows.

Let us construct the function one value at a time. There are m options for $f(1)$. $f(2)$ can take any value except that taken by $f(1)$, and so there are $m-1$ options. $f(3)$ can take any value except those taken by $f(1)$ and $f(2)$, which are two different restrictions, and so there are $m-2$ options. We proceed in this way until we get to $f(n)$, which must be different from the $n-1$ distinct values that came before, leaving $m-(n-1)$ options. Hence the total number of one-to-one functions is $m(m-1)(m-2)\ldots(m-(n-1))$.

Notice that this formula works even when $n > m$, since in that case one of the factors, and hence the entire product, will be 0, showing that there are no one-to-one functions in this case.

(vii) To count the number of onto functions, we first look at the complement - those functions that are not onto. These are the functions for which there is some value in the range - some $1 \leq k \leq m$ - such that $f(x) \neq k$ for all $1 \leq x \leq n$ (i.e. these values are not 'hit' by the function). Let A_k represent those functions that do not hit the value k. We want to count the number of functions in $A_1 \cup A_2 \cup \ldots \cup A_m$ (because the onto functions are those that are *not* in this union). To do so, we will use the inclusion-exclusion formula.

We have $|A_1 \cup A_2 \cup \ldots \cup A_m| = \sum_{\ell=1}^{m} (-1)^{\ell+1} \sum_{I \subset [m], |I|=\ell} |\cap_{i \in I} A_i|$.

Now let $I \subset [m]$ be any subset of ℓ elements; that is, $|I| = \ell$. For a function f to be in $\cap_{i \in I} A_i$, f cannot hit any of the ℓ elements in I. Hence, for each value $f(x)$, there are $m - \ell$ options. Thus it follows that there are $(m-\ell)^n$ such functions; i.e. $|\cap_{i \in I} A_i| = (m-\ell)^n$. As there are $\binom{m}{\ell}$ such sets I of size ℓ, we can evaluate the sum as $|A_1 \cup A_2 \cup \ldots \cup A_m| = \sum_{\ell=1}^{m} (-1)^{\ell+1} \binom{m}{\ell} (m-\ell)^n$.

This was the complement of what we wanted to count. Hence we subtract this from the total number of functions to get the number of onto functions, which gives

$$m^n - \sum_{\ell=1}^{m} (-1)^{\ell+1} \binom{m}{\ell} (m-\ell)^n = \sum_{\ell=0}^{m} (-1)^{\ell} \binom{m}{\ell} (m-\ell)^n.$$

Note: A common mistake is to argue along the following lines: if f is onto, then for every $1 \leq k \leq m$, there is some $1 \leq x \leq n$ with $f(x) = k$. For $k = 1$, there are n options for x with $f(x) = 1$. For $k = 2$, there are $n-1$ options remaining for x with $f(x) = 2$. Proceed in this way until we get to $k = m$, when there will be $n - (m-1)$ options for x with $f(x) = m$. After this, there will be $n - m$ integers x for which we can assign values to $f(x)$; for each of these, there are m options, as they can be assigned freely. Hence the number of such functions is $n \times (n-1) \times \ldots \times (n-(m-1)) \times m \times \ldots \times m = m^{n-m} \prod_{i=0}^{m-1} (n-i)$.

However, this cannot be correct. Indeed, if n is much larger than m, this will be larger than the total number of functions, m^n. For example, if $m = 1$ and n is arbitrary, then there is only 1 possible function (which happens to be onto), but the above formula gives n.

The problem is that functions can be counted multiple times this way, and so the formula overcounts the number of onto functions. This is because for some k, there may be many values x with $f(x) = k$. Thus, in the above construction, we can choose this x in different ways, but end up with the same function in the end.

(viii) We can only have a bijection between sets of the same size, thus the only way a bijection from $[n]$ to $[m]$ is possible is if $n = m$. If $n = m$, then f is a bijection if and only if f is one-to-one, or if and only if f is onto. Thus we obtain the number of bijections by using the formula for the number of one-to-one (or onto) functions in the case $n = m$.

If we use the formula for the number of one-to-one functions, with $n = m$, then we get that the number of bijections from $[n]$ to $[n]$ is $n(n-1)(n-2)\ldots(n-(n-1)) = n!$. (Note that a bijection from $[n]$ to $[n]$ is precisely a permutation, hence the formula $n!$.)

Alternatively, if we use the formula for the number of onto functions, with $n = m$, then the number of bijections is $\sum_{\ell=0}^{n} (-1)^\ell \binom{n}{\ell} (n-\ell)^n$. Note that this gives us the identity

$$n! = \sum_{\ell=0}^{n} (-1)^\ell \binom{n}{\ell} (n-\ell)^n.$$

(ix) We claim that there are two cases: either $f(x)$ is even for all x, or $f(x)$ is odd for all x. Indeed, in either of these cases, $f(x) + f(y)$ will be even for every x and y. If we are not in either of these cases, then there is some x with $f(x)$ even, and some y with $f(y)$ odd. In that case, $f(x) + f(y)$ will be odd, and thus violate the condition. Hence these two cases are the only possible cases.

Consider the first case, when $f(x)$ is even for all x. Thus we are counting the functions $f : [n] \to \{2, 4, \ldots, 2\left\lfloor \frac{m}{2} \right\rfloor\}$. There are $\left\lfloor \frac{m}{2} \right\rfloor$ options for each x, and hence $\left\lfloor \frac{m}{2} \right\rfloor^n$ such functions.

In the second case, $f(x)$ is odd for all x, so we are counting the functions $f : [n] \to \{1, 3, \ldots, 2\left\lfloor \frac{m+1}{2} \right\rfloor - 1\}$. There are $\left\lfloor \frac{m+1}{2} \right\rfloor$ options for each x, and so $\left\lfloor \frac{m+1}{2} \right\rfloor^n$ such functions.

Hence the total number of functions with this property is $\left\lfloor \frac{m}{2} \right\rfloor^n + \left\lfloor \frac{m+1}{2} \right\rfloor^n$.

(x) We assume $m \geq 2$, since if $m = 1$, then $\min_{x \in [n]} f(x) = \max_{x \in [n]} f(x) = 1$. If f satisfies $\max_{x \in [n]} f(x) = \min_{x \in [n]} f(x) + 1$, then there are $m-1$ possible values for $\min_{x \in [n]} f(x)$, namely $1, 2, \ldots, m-1$. Suppose $\min_{x \in [n]} f(x) = k$. Then for each $x \in [n]$, we have $f(x) = k$ or $f(x) = k + 1$.

Hence the function is determined by specifying the value of $k = \min_{x \in [n]} f(x)$ and a set $S \subset [n]$ with $f(x) = k$ for all $x \in S$, and $f(x) = k + 1$ for all $x \notin S$. Note that we cannot have $S = \emptyset$, because otherwise $f(x) = k + 1$ for all x, and $\min_{x \in [n]} f(x) = k + 1$, not k. Similarly, we cannot have $S = [n]$, because then $f(x) = k$ for all x, and $\max_{x \in [n]} f(x) = k$, not $k + 1$. Any other set S would work. Thus there are $2^n - 2$ possible sets for S.

There are $m - 1$ choices for k, and $2^n - 2$ choices for S, and so the number of such functions is $(m-1)(2^n - 2)$.

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